

ANLP: Interim Proposal

Evaluating Multilingual Reasoning through a Chain-of-Thought Lens

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Abstract

We want to answer a simple question: does asking an LLM to “think step by step” help it reason about time, and does the answer change across languages? CoT often improves multi-step reasoning, but multilingual models do not always reason equally well across languages (Zhang et al., 2023; Huang and Chang, 2023; Shi et al., 2023). We will use the Temporal Reasoning Dataset (TRD), which tests temporal tasks in ten languages and includes controlled distractors (Mazzia et al., 2026). We will compare direct answers with CoT in English and Hindi, test whether CoT makes distractors more harmful, try Hinglish, and test cross-lingual alignment prompting. We will also compare the structure of English and Hindi reasoning traces, not just their final answers.

1 The Question

Chain-of-thought (CoT) prompting asks an LLM to generate intermediate steps before giving an answer. Surveys describe CoT as a way to break a difficult problem into smaller pieces, and research shows that asking for explicit steps can improve multi-step reasoning (Zhang et al., 2023; Huang and Chang, 2023). We will use “reasoning trace” to mean the text the model generates; we will not assume that this text is a transparent record of its internal computation (Huang and Chang, 2023).

The language matters too. The MGSM study found that multilingual models can solve reasoning problems in many languages, but English intermediate steps often perform as well as or better than intermediate steps in the language of the question (Shi et al., 2023).

2 Why Temporal Reasoning?

We will use TRD because it gives us problems where the model must manipulate time: date arithmetic, durations, recurrences, and related tasks. TRD generates these problems programmatically,

controls their difficulty, and evaluates them across ten languages (Mazzia et al., 2026).

More importantly, TRD already lets us add controlled distractions. Some distractors relate to time; others have nothing to do with the question, including an example about someone buying a blue motorcycle (Mazzia et al., 2026). This gives us a very direct test of our main idea:

If a model can ignore the motorcycle when it answers directly, does making it think step by step give the motorcycle more chances to enter its reasoning and derail the answer?

TRD evaluates its benchmark without CoT and explicitly leaves the effect of CoT on temporal reasoning for future work (Mazzia et al., 2026). We will investigate that missing condition.

3 What We Will Do

1. Direct answer vs. CoT. For every English/Hindi problem, we will ask for only the final answer and then ask the model to solve the same problem step by step. We will compare accuracy and calculate the *CoT delta*: the change in accuracy caused by turning CoT on. We will also compare the English–Hindi gap across TRD’s difficulty levels.

2. CoT under distraction. We will test each problem with no distractor, a time-related distractor, and an unrelated distractor, both with and without CoT. If a distractor hurts accuracy much more when CoT is on, we will have evidence that the extra reasoning trace increases distraction. We will also check whether the model mentions the distractor in its trace.

3. English, Hindi, and Hinglish. We will ask the model to reason in English, Hindi, and Hinglish. We will treat Hinglish as an exploratory condition:

we want to see whether mixing the two languages changes accuracy or the way the model structures its reasoning, rather than assuming beforehand that it will help.

4. Cross-lingual alignment. Qin et al. introduce Cross-lingual Prompting (CLP), a method where the model first translates or understands a non-English question in English and then solves it step by step (Qin et al., 2023). We will compare regular Hindi CoT with this two-stage approach to see whether using English as an intermediate step improves accuracy or makes Hindi reasoning more similar to English reasoning.

4 How We Will Measure It

Accuracy will be our main metric. We will also convert each CoT trace into a simple sequence of step types (for example, identify a date, add a duration, check a constraint, answer) and compare English and Hindi with normalized edit distance. A lower distance means the two traces have more similar structure; a higher distance means the model takes more different routes. We will also record whether the trace is parseable and whether it mentions a distractor. We will also whether divergence predicts failure, to do this we plan to fit a logistic model $P(\text{err}) \sim \text{RDS} + \text{difficulty} + \text{language}$ and check whether the RDS coefficient is positive and significant. (RDS is the reasoning divergence score, which is the normalised edit distance).

5 What We Expect to Learn

We do not assume that CoT will always help. It should give the model useful space for multi-step calculations, but it may also create more opportunities for irrelevant information to enter the generated reasoning. TRD’s controlled distractors let us test that trade-off directly (Mazzia et al., 2026). The multilingual experiments will tell us whether this trade-off changes across English and Hindi, while CLP will test whether using English as an intermediate step helps the model reason more effectively in Hindi. (Shi et al., 2023; Qin et al., 2023).

6 Model

We will be using the Sarvam-30B model for answer generation (Sarvam Foundation Models Team, 2026). Sarvam-30B is a multilingual large language model designed primarily for Indian languages, with support for both Hindi and English.

Therefore, we can use it to evaluate reasoning in both languages.

7 Measuring Target-Language CoT

Sometimes language models often collapse to English during COT instead of sticking with the target language. Therefore, it is important to measure how often it is doing this behavior to ensure our analysis is strictly on reasoning where the language doesn’t fluctuate a lot.

$$\text{Target-Language COT Ratio} = \frac{N_{\text{Hindi words}}}{N_{\text{Hindi words}} + N_{\text{English words}}}$$

Punctuation and mathematical symbols/numbers won’t be counted towards either language. In order to make the reasoning mostly in Hindi when testing hindi-reasoning, we will be experimenting with probing the model to "Only reason in Devanagri Hindi".

References

- Jie Huang and Kevin Chen-Chuan Chang. 2023. [Towards reasoning in large language models: A survey](#). *arXiv preprint arXiv:2212.10403*.
- Vittorio Mazzia, Sandro Pollastrini, Davide Bernardi, Chiara Rubagotti, and Daniele Amberti. 2026. [Benchmarking multilingual temporal reasoning in llms: The temporal reasoning dataset](#). In *Proceedings of the 16th International Workshop on Spoken Dialogue Systems Technology*, pages 168–181. Association for Computational Linguistics.
- Libo Qin, Qiguang Chen, Fuxuan Wei, Shijue Huang, and Wanxiang Che. 2023. [Cross-lingual prompting: Improving zero-shot chain-of-thought reasoning across languages](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12332–12346. Association for Computational Linguistics.
- Sarvam Foundation Models Team. 2026. Introducing sarvam’s sovereign models. <https://www.sarvam.ai/blogs/sarvam-30b-105b>. Accessed: 2026-03-03.
- Freda Shi, Mirac Suzgun, Markus Freitag, Xuezhi Wang, Suraj Srivats, Soroush Vosoughi, Hyung Won Chung, Yi Tay, Sebastian Ruder, Denny Zhou, Dipanjan Das, and Jason Wei. 2023. [Language models are multilingual chain-of-thought reasoners](#). In *The Eleventh International Conference on Learning Representations*.
- Zhuosheng Zhang, Yao Yao, Aston Zhang, Xiangru Tang, Xinbei Ma, Zhiwei He, Yiming Wang, Mark

Gerstein, Rui Wang, Gongshen Liu, and Hai Zhao. 2023. [Igniting language intelligence: The hitchhiker’s guide from chain-of-thought reasoning to language agents](#). *arXiv preprint arXiv:2311.11797*.